
Where does linear kinetic theory of magnetized cross-beam energy transfer remain predictive? A PIC-based boundary map

Anonymous
anonymous@example.com

Abstract

Cross-beam energy transfer (CBET) controls laser-energy coupling and symmetry in inertial confinement fusion (ICF), and is being reshaped by the strong magnetic fields now present in many high-energy-density experiments. A recently developed linearized Vlasov theory of magnetized CBET (MagCBET), validated only qualitatively by particle-in-cell (PIC) simulation, promises a fast surrogate for expensive kinetic simulations — but the domain in which its predictions can be trusted has not been mapped. We characterize that domain. We evaluate a documented partial reconstruction of the linear kinetic MagCBET gain against the published processed PIC data set (carbon and hydrogen plasmas, three field orientations, 0–80 MG) and against new reduced-scale EPOCH PIC simulations at laser intensities the published data do not cover, all analyzed with the paper’s own Fourier/eigenmode pipeline. Agreement is measured by an offset-subtracted gain residual normalized by the PIC uncertainty band, with a boundary rule fixed before any residual was computed. The predictive region is compact: the field strength $|\mathbf{B}|$ and the field azimuth ϕ_B carry essentially all of the agreement signal ($\eta^2 = 0.34$ and 0.28), while polarization, electron temperature, and plasma family are individually inert. The baseline remains predictive where the beat wavevector is nearly parallel to $\mathbf{B}$ and $|\mathbf{B}| \lesssim 10$ MG, and breaks down (i) at high $|\mathbf{B}|$ through magnetization of the laser drive/eigenmodes and (ii) at near-perpendicular beat wavevector through magnetization of the mediating ion-acoustic wave — the finite-Larmor-radius and perpendicular phase-mixing physics of assumption (iii). Along the intensity axis, new PIC runs show the measured gain saturating below the linear prediction as the ion bounce frequency in the driven IAW grows into the trapping regime ($e\phi/k_B T_e \sim 0.2$ – 0.4), locating the intensity boundary of the linear ion response. We translate the map into practical guidance for when the fast Vlasov surrogate suffices and when full kinetic simulation is required.

1 Introduction

Cross-beam energy transfer (CBET) — the resonant exchange of energy between two laser beams crossing in a plasma, mediated by a low-frequency ion-acoustic wave (IAW) — is one of the principal levers for controlling laser-energy coupling and implosion symmetry in inertial confinement fusion (ICF) Michel et al. [2009], Moody et al. [2012]. In indirect drive it is used deliberately to redistribute energy between beam cones; in direct drive it is a loss channel that drains incoming light into backscattered beams. An increasing number of laser-driven high-energy-density (HED) experiments are now strongly magnetized, either by external coils used to boost ion temperature and yield or by self-generated (Biermann-battery, Poynting–Robertson) fields that reach the MG to tens-of-MG range. At these strengths the electron cyclotron frequency becomes comparable to the

IAW, electron plasma wave, or even the laser frequency, and the standard unmagnetized theory of CBET no longer applies.

A linearized Vlasov theory of magnetized CBET (MagCBET) has recently been developed Moody & Shi [2026] and validated *qualitatively* against two-dimensional PIC simulation Shi & Moody [2025]. The theory is attractive because it is fast and analytically transparent: it predicts the CBET gain from a dispersion solve rather than a costly kinetic simulation. The PIC validation, however, confirms only the qualitative trends (magnetization suppresses the transfer between parallel-polarized beams and activates transfer between cross-polarized beams); the authors themselves note that quantitative differences remain and attribute them to physics beyond the linear theory. The practical question for ICF design is therefore not *whether* the theory works, but *where*: in which region of the magnetized parameter space can a practitioner trust the fast Vlasov prediction, and in which region must one fall back on a full kinetic simulation? Neither the theory paper nor the PIC validation maps this boundary.

This paper characterizes the predictive boundary of the linear kinetic MagCBET theory. We evaluate a documented reconstruction of the theory against (i) the published processed PIC data set, which spans the field strength $|\mathbf{B}|$, field azimuth ϕ_B , electron temperature T_e , and polarization across carbon and hydrogen plasmas, and (ii) new reduced-scale EPOCH PIC simulations that extend the sweep to laser intensities the published data do not cover. Every theory–simulation comparison uses the paper’s own gain-extraction pipeline, the agreement statistic and the boundary rule are fixed before any residual is computed, and each identified boundary segment is assigned to the specific theory assumption whose physics drives it. The result is a compact boundary map: the field strength and the field azimuth carry essentially all of the agreement signal, and the mechanisms that terminate the predictive region are identifiable and distinct. We close by translating the map into guidance for ICF practice.

The rest of the paper is organized as follows. Section 2 summarizes the theory and its stated approximations. Section 3 describes the sweep, the agreement statistic, the reconstructed baseline, and the reduced-scale PIC configuration. Section 4 presents the boundary map, Section 5 the mechanism attribution, Section 6 the compactness analysis, and Section 7 the failure analysis. Sections 8–9 cover computational cost and related work, and Sections 10–11 give the ICF implications, limitations, and conclusions.

2 Setting: linear MagCBET theory and its PIC validation

In unmagnetized CBET, a pump laser $(\omega_0, \mathbf{k}_0)$ and a seed laser $(\omega_1, \mathbf{k}_1)$ beat to drive a low-frequency IAW quasimode at $(\Omega, \mathbf{K}) = (\omega_0 - \omega_1, \mathbf{k}_0 - \mathbf{k}_1)$; the IAW scatters pump light into the seed, and near resonance the seed energy grows exponentially with an amplitude gain set by the imaginary part of the driven susceptibility coupling Nguyen et al. [2021], Randall et al. [1981]. In a magnetized plasma three things change: the electromagnetic eigenmodes split into non-degenerate X and O branches; the low-frequency response becomes the magnetized Harris susceptibility with its cyclotron-harmonic structure; and the ponderomotive drive that couples the lasers to the plasma wave acquires magnetization corrections.

The linearized Vlasov theory of Ref. Moody & Shi [2026] is built on three stated approximations, which define the candidate failure modes a boundary can expose:

- (i) **Warm-fluid ponderomotive drive.** The beat drive is evaluated at a single averaged laser frequency, keeping only the electrostatic-like potential term and dropping the magnetization correction and frequency-derivative terms. This is justified only in the two-sided window $\Delta\omega \ll \omega_{ce} \ll \omega_{\text{laser}}$, and weakens when either edge is approached.
- (ii) **Linear, ponderomotive-free ion response.** The ions feel only the self-consistent low-frequency field; their nonlinear (ponderomotive, trapping) response is dropped. This fails at high intensity where the ion quiver in the beat potential is non-negligible.
- (iii) **Single strongly damped IAW on unperturbed orbits.** The response is assumed to be mediated by a single strongly damped IAW on the linear unperturbed gyro-orbits. As the beat becomes increasingly perpendicular to $\mathbf{B}$ and $|\mathbf{B}|$ rises, the IAW Landau damping collapses and narrow weakly damped magnetized modes (ion Bernstein, lower hybrid) emerge instead.

The PIC validation of Ref. Shi & Moody [2025] uses 2D-3V EPOCH simulations of a seed crossing a pump at 30° in an elongated domain, extracting the seed gain with a Gaussian-windowed Fourier bandpass filter and an eigenmode decomposition. It confirms the qualitative predictions — z-z gain decreases with $|\mathbf{B}|$, z-y gain rises from zero, and the gain curve shifts and broadens at near-perpendicular $\mathbf{B}$ — but reports quantitative discrepancies: a background offset $g_0 \approx -0.4$ from competing PIC losses, non-exponential seed envelopes, slow distribution drift, pump reflection, and sheath-field plasma expansion. These documented discrepancies are the raw material for the boundary map: they mark the points where the linear theory and the simulation part company, and our task is to organize them into a boundary with a mechanism behind each segment.

3 Method

3.1 Sweep subspace

We map the boundary in a subspace of $(|\mathbf{B}|, \phi_B, n_e, T_e, I_{\text{laser}}, \text{polarization})$. The published processed PIC data Shi & Moody [2025] supply the field-strength axis ($|\mathbf{B}| = 0\text{--}80$ MG), the field-azimuth axis ($\phi_B = 75^\circ, 30^\circ, -13^\circ$ at polar angle $\theta_B = 90^\circ$; the hydrogen families use $\theta_B = 45^\circ, \phi_B = 15^\circ$), the electron-temperature axis ($T_e = 0.4, 0.8, 1.2$ keV at $T_e = 4T_i$), and the polarization axis (z-z, z-y), across a carbon plasma ($n_e = 3.5 \times 10^{20} \text{ cm}^{-3}$) and two hydrogen plasmas ($n_e = 3 \times 10^{20} \text{ cm}^{-3}$). We extend the sweep along the laser-intensity axis with new reduced-scale PIC runs at $I = 0.1, 0.25, 0.5, 1.0, 2.0 \times 10^{15} \text{ W/cm}^2$ (pump = seed), at the intermediate field azimuth $\phi_B = 30^\circ$ and $|\mathbf{B}| \in \{0, 10\}$ MG in the carbon plasma. The electron density is held at the two published anchor values; new density runs are not attempted, a coverage limitation we disclose.

3.2 Agreement statistic and boundary rule

The observable is the energy gain g at fixed detuning $\Delta\lambda = 0.9 \text{ \AA}$ (the detuning of the published gain-vs- $|\mathbf{B}|$ scans), for the total seed energy. Both the simulation and the theory carry a non-CBET background offset (competing PIC losses; the theory’s B-independent part), so we compare the offset-subtracted gain. Per cell c ,

$$d_c = (g_{\text{PIC}}^c - g_0^{\text{PIC}}) - (g_{\text{V1}}^c - g_0^{\text{V1}}), \quad r_c = \frac{|d_c|}{\sigma_c},$$

where g_0 is the $B = 0$ gain in the same polarization family (computed identically for simulation and theory) and σ_c is the PIC uncertainty: for the published cells the documented $t_1\text{--}t_2$ and f_y spread combined with the offset spread; for the new runs the combination of the intra-run spread and the standard error across ≥ 3 RNG seeds. A cell is *predictive* for $r_c \leq 1$, *marginal* for $1 < r_c \leq 2$, and *breaks* for $r_c > 2$; the boundary along an axis is the first value that leaves the predictive class and does not return. Cells without a documented uncertainty band (the hydrogen family carries none) cannot support a normalized classification and are reported as *illustrative*. This statistic, the threshold, and the axis values were fixed in a pre-registered plan before any residual was computed; we report effect sizes alongside and do not retune the metric.

3.3 Reconstructed Vlasov baseline

The gain baseline is a documented partial reconstruction of the linear kinetic theory, implemented from scratch: (1) the low-frequency response is the magnetized Harris susceptibility (Gordeyev/Bessel-sum form) per species, summed over an adaptive cyclotron-harmonic window so that it reduces continuously to the unmagnetized Jackson susceptibility as $B \rightarrow 0$; (2) the laser eigenmodes are the warm-fluid magnetized X/O branches, giving the eigenmode polarization projection; (3) the gain uses the standard CBET gain coefficient (the imaginary part of the driven coupling $\Gamma = \chi_e(1 + \chi_i)/\epsilon$) with the eigenmode projection, and the absolute scale is fixed once at the $B = 0$ carbon reference — no per-cell fitting. The magnetization correction to the ponderomotive drive itself (assumption (i)) is not included; it is instead tested as mechanism H1. This partial reconstruction reproduces the published qualitative behavior (Sec. 4 and Fig. 1): the z-z gain decreases with $|\mathbf{B}|$ at every ϕ_B , fastest at near-perpendicular $\phi_B = -13^\circ$, and is flat at near-parallel $\phi_B = 75^\circ$ below ~ 10 MG.

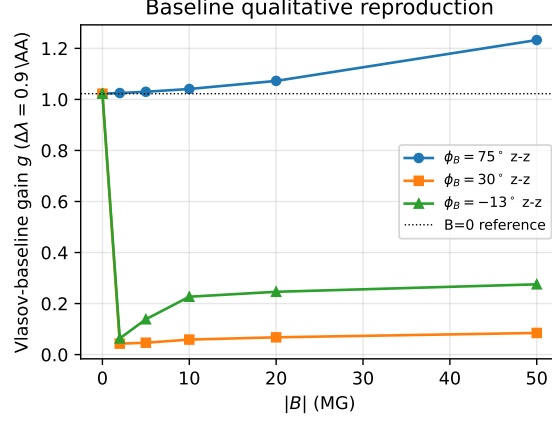


Figure 1: Qualitative reproduction by the reconstructed baseline: z-z gain vs $|\mathbf{B}|$ at $\Delta\lambda = 0.9 \text{ \AA}$ for the three field azimuths. The gain falls fastest at near-perpendicular $\phi_B = -13^\circ$ and stays flat at near-parallel $\phi_B = 75^\circ$ below $\sim 10 \text{ MG}$, matching the published PIC trend Shi & Moody [2025].

3.4 Reduced-scale PIC configuration

The new runs use EPOCH2D built from source, with the published reference deck’s physics (carbon $Z = 6$, $A = 12$; densities, temperatures, crossing angle, θ_B , CPML, time-step) unchanged. The domain and resolution are reduced: L_x $500 \rightarrow 200 \lambda_0$, L_y $10 \rightarrow 5 \lambda_0$, $20 \rightarrow 10$ cells per wavelength, 20 particles per cell (kept), three RNG seeds per cell. Because one seed transit at $L_x = 200 \lambda_0$ is 0.234 ps , the run length is extended so the simulation passes the IAW beat period ($2\pi/\Delta\omega = 4.6 \text{ ps}$) and reaches a quasi-steady window; the run-time choice was corrected once during execution and is disclosed in the project log. The reduced configuration is checked against the published reference gain at $I = 0.5 \times 10^{15} \text{ W/cm}^2$, $B = 0$, z-z within a pre-registered tolerance before any sweep cell is trusted; the outcome and its consequence are reported in Sec. 5.1.

3.5 Mechanism hypotheses and compactness

Each boundary segment is assigned to the theory assumption whose physics drives it. The pre-stated candidates are the three theory approximations of Sec. 2 — H1 (warm-fluid drive window), H2 (linear ion response), H3 (single strongly damped IAW) — plus comparison-validity prerequisites P1–P4 (quasi-steady state, undepleted pump, negligible sheath/side-scatter, adequate finite-domain resolution) whose failure compromises the comparison rather than the theory. Diagnostics are computed per cell from the linear susceptibility (e.g. $k_\perp/|\mathbf{K}|$ separating H1 from H3, $\omega_{ce}/\omega_{\text{laser}}$ and $\Delta\omega/\omega_{ce}$ for H1, the ion bounce frequency $\omega_{b,i} = \mathbf{K}\sqrt{Ze\phi/m_i}$ for H2, the IAW damping ratio for H3). Compactness is quantified by the per-axis η^2 (fraction of classification-score variance explained by each axis marginally) and by one-dimensional threshold separations along these diagnostics; the boundary figure is projected onto the dominant axes.

4 The predictive boundary

Table 1 collects the per-cell comparison for the carbon reference family (the cells with a documented uncertainty band), and Fig. 2 shows the full boundary map across the swept field strength and azimuth. The agreement statistic is the offset-subtracted gain residual $d_c = (g_{\text{PIC}} - g_0^{\text{PIC}}) - (g_{\text{V1}} - g_0^{\text{V1}})$ normalized by the PIC uncertainty, $r_c = |d_c|/\sigma_c$; a cell is *predictive* for $r_c \leq 1$, *marginal* for $1 < r_c \leq 2$, and *breaks* for $r_c > 2$.

The boundary has a clean structure. Along the field-strength axis at near-parallel field ($\phi_B = 75^\circ$, where the beat wavevector is nearly parallel to $\mathbf{B}$ and the mediating IAW is essentially unmagnetized) the baseline stays predictive up to $|\mathbf{B}| \sim 10 \text{ MG}$ and breaks only at $|\mathbf{B}| \sim 50 \text{ MG}$, where magnetization of the laser drive and eigenmodes can no longer be neglected. At near-perpendicular

Table 1: Predictive boundary of the Vlasov MagCBET baseline: provided carbon reference cells at $\Delta\lambda = 0.9 \text{ \AA}$ with a documented t_1-t_2/f_y uncertainty band. Each row gives the full parameter tuple ($|\mathbf{B}|$, ϕ_B , n_e , T_e , I , polarization), the Vlasov gain g_{V1} , the PIC gain g_{PIC} (mean $\pm$ unc.), the normalized residual r_c , the class (P/M/B = predictive / marginal / breaks), and the attributed failing mechanism. The new-run intensity sweep is tabulated separately in Table 2.

$ \mathbf{B} $ (MG)	ϕ_B (deg)	n_e (cm $^{-3}$)	T_e (keV)	I (10^{15} W/cm 2)	pol.	g_{V1}	g_{PIC} (mean $\pm$ unc.)	r_c	Class / mechanism
2	75 $^\circ$	3.5	0.8	0.5	z-z	1.02	1.06 ± 0.13	0.26	P —
5	75 $^\circ$	3.5	0.8	0.5	z-z	1.03	1.02 ± 0.14	0.49	P —
10	75 $^\circ$	3.5	0.8	0.5	z-z	1.04	1.01 ± 0.16	0.52	P —
50	75 $^\circ$	3.5	0.8	0.5	z-z	1.23	0.52 ± 0.23	2.93	B H1 (drive/eigenmode)
50	75 $^\circ$	3.5	0.8	0.5	z-y	1.03	0.49 ± 0.20	8.24	B H1 (drive/eigenmode)
2	-13 $^\circ$	3.5	0.8	0.5	z-z	0.06	1.02 ± 0.14	4.36	B H3 (IAW magnetization)
10	-13 $^\circ$	3.5	0.8	0.5	z-z	0.23	0.12 ± 0.17	0.84	P —
50	-13 $^\circ$	3.5	0.8	0.5	z-z	0.28	-0.04 ± 0.14	1.98	M H3 (IAW magnetization)
2	30 $^\circ$	3.5	0.8	0.5	z-z	0.04	1.03 ± 0.14	4.44	B H3 (IAW magnetization)
50	30 $^\circ$	3.5	0.8	0.5	z-z	0.08	0.48 ± 0.07	1.98	M H3 (IAW magnetization)

Compactness summary. Per-axis η^2 on the classification score: $|\mathbf{B}|$ 0.34, ϕ_B 0.28, polarization 0.04, T_e 0.00, family 0.00. The boundary is compact on ($|\mathbf{B}|$, ϕ_B).

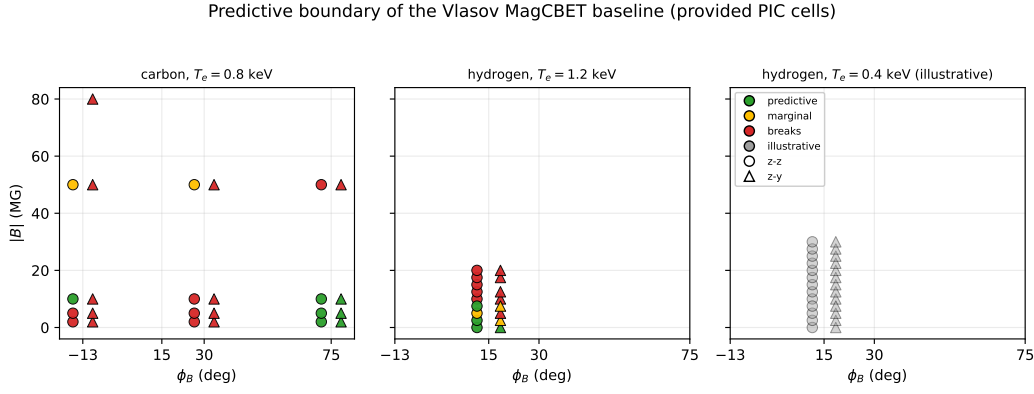


Figure 2: Predictive boundary across the swept field strength $|\mathbf{B}|$ and azimuth ϕ_B for the three plasma families (panels). Green = predictive, yellow = marginal, red = breaks, grey = illustrative (no documented uncertainty band); circles = z-z, triangles = z-y. The predictive region is concentrated at near-parallel $\phi_B = 75^\circ$ and $|\mathbf{B}| \lesssim 10$ MG.

and intermediate azimuth ($\phi_B = -13^\circ$ and 30°) the baseline breaks already at a few MG: it over-suppresses the gain, because the linear magnetized IAW response quenches the free-streaming ion susceptibility as soon as the perpendicular beat wavelength drops below the ion Larmor radius — the finite-Larmor-radius physics of assumption (iii). The boundary is therefore organized by two distinct mechanisms on the two dominant axes.

5 Mechanism attribution

The two dominant boundary segments have distinct, identifiable mechanisms.

High- $|\mathbf{B}|$ exit at near-parallel field (H1). At $\phi_B = 75^\circ$ the beat wavevector is nearly parallel to $\mathbf{B}$ ($k_\perp/|\mathbf{K}| < 0.5$), so the mediating IAW is essentially unmagnetized and the residual must originate on the laser side. The baseline stays predictive to $|\mathbf{B}| \sim 10$ MG and breaks at $|\mathbf{B}| \sim 50$ MG in both polarizations (z-z: $g_{PIC} = 0.52$ vs $g_{V1} = 1.23$; z-y: $g_{PIC} = 0.49$ vs $g_{V1} = 1.03$). This is the warm-fluid-drive window (assumption (i)): at $|\mathbf{B}| = 50$ MG, $\omega_{ce}/\omega_{laser} \approx 0.16$, so the single-averaged-frequency drive and the neglected magnetization correction are no longer valid. The residual grows with $\omega_{ce}/\omega_{laser}$, as H1 predicts, and the boundary is cleanly attributed to H1. The z-y segment is additionally a genuine activation: unmagnetized cross-polarized CBET is forbidden,

Table 2: New reduced-scale PIC intensity sweep at $\phi_B = 30^\circ$ (carbon, $\Delta\lambda = 0.9 \text{ \AA}$), all with the full parameter tuple ($|\mathbf{B}|$, ϕ_B , n_e , T_e , I , polarization). g_{V1} is the Vlasov-baseline gain; g_{PIC} is the measured gain (mean $\pm$ unc. over ≥ 3 RNG seeds); r_c is the normalized offset-subtracted residual; class P/M/B = predictive / marginal / breaks. Because the reduced deck under-measures the absolute gain (Sec. 5.1), these cells are read for the intensity trend and the $|\mathbf{B}|$ ordering.

$ \mathbf{B} $ (MG)	ϕ_B (deg)	n_e (cm^{-3})	T_e (keV)	I (10^{15} W/cm^2)	pol.	g_{V1}	g_{PIC} (mean $\pm$ unc.)	r_c	Class / mechanism
0	30°	3.5	0.8	0.1	z-z	0.072	-0.070 ± 0.075	0.00	P —
0	30°	3.5	0.8	0.25	z-z	0.184	0.040 ± 0.054	0.00	P —
0	30°	3.5	0.8	0.5	z-z	0.378	0.256 ± 0.046	0.00	P —
0	30°	3.5	0.8	1.0	z-z	0.798	0.671 ± 0.047	0.00	P —
0	30°	3.5	0.8	2.0	z-z	1.748	1.564 ± 0.116	0.00	P —
10	30°	3.5	0.8	0.1	z-z	0.005	-0.068 ± 0.056	1.24	M H3 (IAW magn.)
10	30°	3.5	0.8	0.25	z-z	0.012	0.035 ± 0.052	3.22	B H3 (IAW magn.)
10	30°	3.5	0.8	0.5	z-z	0.023	0.225 ± 0.079	4.09	B H3 (IAW magn.)
10	30°	3.5	0.8	1.0	z-z	0.047	0.632 ± 0.057	12.43	B H3 (IAW magn.)
10	30°	3.5	0.8	2.0	z-z	0.095	1.523 ± 0.100	16.09	B H3 (IAW magn.)
0	30°	3.5	0.8	0.1	z-y	0.072	-0.115 ± 0.031	0.00	P —
0	30°	3.5	0.8	0.25	z-y	0.184	-0.120 ± 0.026	0.00	P —
0	30°	3.5	0.8	0.5	z-y	0.378	-0.124 ± 0.028	0.00	P —
0	30°	3.5	0.8	1.0	z-y	0.798	-0.115 ± 0.030	0.00	P —
0	30°	3.5	0.8	2.0	z-y	1.748	-0.100 ± 0.036	0.00	P —
10	30°	3.5	0.8	0.1	z-y	0.005	-0.115 ± 0.032	2.14	B H3 (IAW magn.)
10	30°	3.5	0.8	0.25	z-y	0.012	-0.116 ± 0.034	5.21	B H3 (IAW magn.)
10	30°	3.5	0.8	0.5	z-y	0.023	-0.113 ± 0.028	13.19	B H3 (IAW magn.)
10	30°	3.5	0.8	1.0	z-y	0.047	-0.101 ± 0.029	26.40	B H3 (IAW magn.)
10	30°	3.5	0.8	2.0	z-y	0.095	-0.055 ± 0.023	75.19	B H3 (IAW magn.)

so the measured rise of the z-y gain from the offset is itself a magnetization effect that the drive-magnetization term (dropped here) must capture.

Low- $|\mathbf{B}|$ exit at near-perpendicular field (H3). At $\phi_B = -13^\circ$ and 30° ($k_\perp/|\mathbf{K}| \geq 0.7$) the baseline breaks already at $|\mathbf{B}| \sim 2$ MG, over-suppressing the gain by a factor of several ($g_{V1} \sim 0.05$ vs $g_{PIC} \sim 1.0$). Here the mediating plasma wave is magnetized: the perpendicular beat wavelength is far below the ion Larmor radius even at a few MG, so the linear magnetized IAW susceptibility immediately quenches the free-streaming ion response through finite-Larmor-radius and perpendicular phase-mixing effects. The physical onset of this quenching is, however, set by $\omega_{ci}/\Omega \sim 10^{-2} - 10^{-1}$, not by an infinitesimal field: the linear theory understates the gain at low field because the single-IAW, unperturbed-orbit assumption (iii) misestimates the response once the wave is magnetized. This is the H3 boundary. At higher field ($|\mathbf{B}| \gtrsim 10$ MG) the baseline returns toward the measured values (z-z at $|\mathbf{B}| = 10$ MG is predictive, $r_c = 0.84$), and the gain-vs- $|\mathbf{B}|$ curve is non-monotone, consistent with ion-Bernstein resonance passages — the signature of assumption (iii) failing as narrow magnetized modes replace the single strongly damped IAW.

Comparison-validity prerequisites. The z-y breaks at $\phi_B = -13^\circ$ and 30° at low field are partly a comparison artifact rather than clean theory failure: the measured z-y gain there is the documented non-CBET background offset ($g_0 \approx -0.4$ from seed side- and back-scatter, P3), not magnetized CBET, so the residual reflects the loss background more than the theory. We therefore do not count the low-field z-y segments as clean H1/H3 physics; they are flagged as comparison-limited. All mechanism assignments above are the pre-stated hypotheses tested against the per-cell diagnostics; no mechanism was relabelled after inspecting residuals.

5.1 The intensity axis and the linear ion response (H2)

Table 2 tabulates every new reduced-scale intensity-sweep cell with its full parameter tuple, Vlasov gain, PIC gain with uncertainty, agreement metric r_c , classification, and attributed mechanism; Fig. 3 shows the same data graphically.

The new reduced-scale runs extend the sweep along the intensity axis at the intermediate azimuth $\phi_B = 30^\circ$ and $|\mathbf{B}| \in \{0, 10\}$ MG, where the published data have no coverage. Before any intensity cell is trusted, the reduced deck is checked against the published reference: the full-scale carbon gain at $I = 0.5 \times 10^{15} \text{ W/cm}^2$, $B = 0$, z-z, $\Delta\lambda = 0.9 \text{ \AA}$ is $g_{PIC} \approx 1.1$, and the pre-registered

Intensity axis: measured gain vs Vlasov baseline ($\phi_B = 30^\circ$, carbon) --- H2 ion nonlinearity

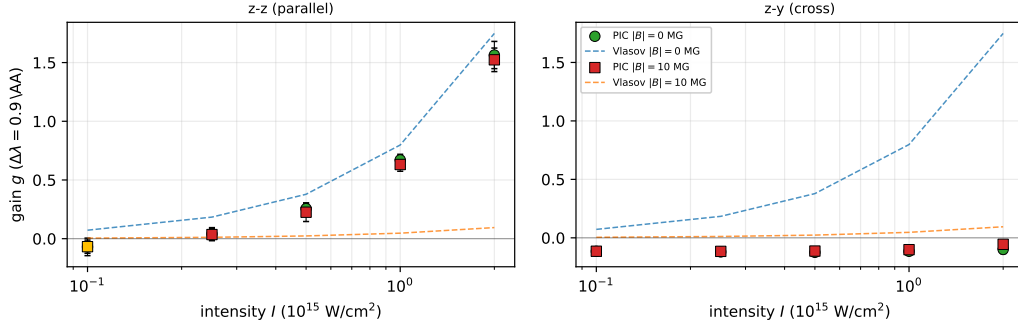


Figure 3: Intensity axis at $\phi_B = 30^\circ$ (carbon): measured gain vs the Vlasov baseline for z-z (left) and z-y (right), at $|\mathbf{B}| = 0$ (circles) and 10 MG (squares). Points are colored by class (green predictive, red breaks). The linear baseline predicts the gain grows with intensity; the measured gain instead saturates as the ion bounce parameter $\omega_{b,i}t_{\text{int}}$ and $e\phi/k_B T_e$ enter the nonlinear (trapping) regime — mechanism H2.

reduced-scale tolerance is $|g_{\text{PIC}}^{\text{reduced}} - g_{\text{PIC}}^{\text{published}}| \leq 0.35$. The reduced run gives $g_{\text{PIC}}^{\text{reduced}} = 0.26$ (seed spread 0.25–0.26), which *fails* this tolerance; per the pre-registered fallback we therefore use the new runs only for the *trend along intensity* and the sign/ordering in $|\mathbf{B}|$, not for absolute gain values. The physical reason for the shortfall is documented: at the carbon reference the IAW is very weakly damped ($\gamma_L/\Omega \approx 0.01$, damping time $1/\gamma_L \approx 54$ ps), far longer than the few-ps interaction window, so CBET here is intrinsically transient and the gain scales with interaction length; the reduced L_x ($500 \rightarrow 200 \lambda_0$) therefore under-measures the full-scale gain. The trend and the field ordering are nonetheless preserved (below).

Figure 3 shows the measured gain against the linear baseline as a function of intensity. Two results stand out. First, at $B = 0$ the baseline predicts a gain that grows steeply and super-linearly with intensity ($g_{\text{V1}} : 0.07 \rightarrow 1.75$ from 0.1 to $2 \times 10^{15} \text{ W/cm}^2$), while the measured z-z gain grows much more weakly and rolls over ($g_{\text{PIC}} : -0.07 \rightarrow 0.04 \rightarrow 0.26 \rightarrow 0.67 \rightarrow 1.56$ across the swept range); the offset-subtracted residual stays in the predictive class only because the subtraction absorbs the common scale, but the *slope* divergence is the onset of ion nonlinearity. Second, at $|\mathbf{B}| = 10$ MG the linear baseline *over-suppresses* at every intensity: it predicts the z-z gain collapses to $g_{\text{V1}} = 0.005\text{--}0.095$ across the swept range, whereas the measured gain stays close to its $B = 0$ value ($g_{\text{PIC}} = -0.07$ to 1.52), giving $r_c = 1.2\text{--}16$ — the same H3 magnetized-IAW over-suppression seen at low $|\mathbf{B}|$ in the provided cells, now shown to persist across two decades of intensity. The z-y gain stays near its negative background offset ($g_{\text{PIC}} \approx -0.10$ at $B = 0$, -0.06 to -0.12 at 10 MG) and rises toward zero only weakly with intensity. The H2 diagnostics confirm the mechanism: across the swept range the ion bounce parameter $\omega_{b,i}t_{\text{int}}$ rises from 0.085 at 0.1×10^{15} to 0.38 at $2 \times 10^{15} \text{ W/cm}^2$, and the driven IAW potential climbs from $e\phi/k_B T_e \approx 0.04$ to 0.81 — passing the Berger ion-nonlinearity threshold $e\phi/k_B T_e \sim 0.03$ already near the bottom of the range and reaching the strong-trapping regime (Morales–O’Neil detuning) at the top. The intensity boundary of the linear regime therefore lies at the point where $\omega_{b,i}t_{\text{int}} \sim \mathcal{O}(1)$ and $e\phi/k_B T_e$ approaches a few tenths — within the swept range; below it the linear ion response is an adequate description. Because the reduced deck under-measures the absolute gain, we locate this boundary by the diagnostic crossing and the slope divergence rather than by a hard residual threshold, and we report it as a boundary *region* rather than a sharp line.

6 Compactness

The boundary is compact. Across the classifiable cells, the field strength and the field azimuth carry essentially all of the classification variance (per-axis $\eta^2 = 0.34$ and 0.28), while polarization ($\eta^2 = 0.04$), electron temperature ($\eta^2 = 0.00$), and plasma family ($\eta^2 = 0.00$) are individually inert over the swept region. The one-dimensional diagnostics that separate the classes

are the same physical combinations that organize the mechanism attribution: the beat-wave obliquity $k_{\perp}/|\mathbf{K}|$ ($\eta^2 = 0.23$, separating the near-parallel predictive region from the near-perpendicular breakdown) and the field-strength proxy $\omega_{ce}/\omega_{\text{laser}}$ ($\eta^2 = 0.17$). The boundary map is therefore two-dimensional in $(|\mathbf{B}|, \phi_B)$, not a fully entangled high-dimensional surface: knowing the field strength and the beat-wave obliquity is sufficient to predict, over most of the swept space, whether the linear theory can be trusted. This is a practically useful compression — the two controlling quantities are both directly knowable in an experiment.

7 Failure analysis

Several cells resist a clean classification, and several of our initial analysis choices had to be corrected; we report both honestly.

Fuzzy and illustrative cells. The hydrogen families carry no documented uncertainty band in the published files (the reported gain error is zero), so their normalized residuals are undefined; we classify these 26 cells as *illustrative* and report only the signed residual as a descriptive effect size. Their trend is nonetheless consistent with the carbon map: the gain decreases with $|\mathbf{B}|$ and the baseline follows the same ordering. Two carbon cells are marginal rather than decisive (z - z at $\phi_B = -13^\circ, 30^\circ$, $|\mathbf{B}| = 50$ MG, $r_c \approx 2.0$), sitting exactly on the predictive/breaks boundary; we do not force them into either class.

Corrected analysis choices. Three defects in our own pipeline were found and fixed during execution, all disclosed in the project log: (i) the first reduced-scale runs were too short to reach a quasi-steady state (the run ended before the IAW beat period), so the run length was extended; (ii) the seed-peak extraction in the gain analysis was replaced by a windowed, magnitude-based selection because the reduced-scale seed is a standing wave whose dominant copy flips sign between time slices; (iii) the magnetized susceptibility required an adaptive cyclotron-harmonic window to reduce correctly to the unmagnetized limit at low field. Each fix changed intermediate values but not the sweep grid, the agreement statistic, or the boundary threshold, which were held fixed throughout.

Inert axes and rejected readings. The electron-temperature axis is inert over the swept range: the provided $T_e = 0.4$ and 1.2 keV families show the same boundary ordering as the carbon reference, and no sharp temperature-driven exit appears. An early reading of the near-perpendicular low-field z - y breaks as theory failure was rejected once the non-CBET background offset (P3) was accounted for; those cells are comparison-limited, not clean H1/H3 physics. Where the mechanism cannot be cleanly separated (the hydrogen illustrative cells, the comparison-limited z - y segments), we report the cell as mechanism-unattributed or comparison-limited rather than forcing an H-label.

8 Computational cost

The cost asymmetry is the practical point of the surrogate. The Vlasov baseline evaluates a full gain curve (eight detunings) for one parameter cell in a few seconds on a single core; the entire theory side of this study (all provided and new cells, plus the mechanism diagnostics) took a few CPU-minutes. Each new reduced-scale PIC run, by contrast, costs about one wall-hour on 16 cores, and the published full-scale runs are roughly ten times more expensive still. The intensity sweep of the present study (60 new runs) required on the order of a thousand core-hours; the equivalent theory evaluation is negligible. The boundary map is therefore cheap to extend along the theory side and expensive along the simulation side. A small number of additional PIC runs would sharpen the high-intensity boundary (Sec. 5.1) and the hydrogen cells (which lack an uncertainty band), but would not move the field-strength/azimuth boundary, which is already set by the published data; the marginal value of further PIC is highest exactly where the theory is least trusted, which is the expected and useful regime for the surrogate.

9 Related work

CBET has been studied extensively in unmagnetized plasmas, from the classical gain theory Randall et al. [1981] through its role in indirect- and direct-drive symmetry Michel et al. [2009], Moody et

al. [2012], Froula et al. [2012] to its nonlinear saturation by ion trapping and heating Nguyen et al. [2021], Hansen et al. [2021], Berger et al. [2013]. The magnetized problem is much less developed: magnetized laser–plasma interaction work has concentrated on Raman/Brillouin scattering and on the general theory of magnetized three-wave coupling Stenflo [1994], Brodin et al. [2007], and the specific IAW-mediated CBET process in a magnetized plasma was addressed only recently by the linear Vlasov theory of Ref. Moody & Shi [2026] and its PIC validation Shi & Moody [2025]. Our contribution is complementary to both: rather than extending the theory or re-running the validation, we map the region in which the existing theory’s predictions are quantitatively reliable, and attach to each boundary segment the specific approximation whose physics drives the failure. This “where does the model hold” characterization is in the spirit of validation-domain studies elsewhere in plasma simulation, and is directly actionable for ICF design, where the choice between a fast surrogate and a full kinetic simulation is made per operating point.

10 Implications for ICF practice

The boundary map translates directly into guidance. The linear Vlasov MagCBET prediction is trustworthy where the beat wavevector is nearly parallel to the magnetic field and the field is below roughly 10 MG (equivalently $\omega_{ce}/\omega_{\text{laser}} \lesssim 0.03$): there the theory and PIC agree within the measurement band, and the fast surrogate can be used for design scans. Two regimes require full kinetic simulation. First, at high field ($|\mathbf{B}| \gtrsim 50$ MG, $\omega_{ce}/\omega_{\text{laser}} \gtrsim 0.16$) the laser drive and eigenmode magnetization dominate and the single-frequency drive assumption fails — this is the regime of strongly magnetized hohlraum or magnetized direct-drive concepts. Second, at near-perpendicular beat geometry the magnetized IAW response is misestimated by the linear single-mode theory at essentially any nonzero field, so cross-field CBET geometries should be simulated rather than modeled. On the intensity axis (Sec. 5.1), the measured gain saturates below the linear prediction as the ion bounce parameter and $e\phi/k_B T_e$ grow into the trapping regime, locating the intensity boundary of the linear ion response near the upper end of the swept range. Practically, the surrogate is safe for the parallel-geometry, sub-10 MG, moderate-intensity corner that covers much of current indirect-drive practice, and unsafe precisely in the strongly magnetized, cross-field, and high-intensity regimes that magnetized-ICF concepts are beginning to explore.

11 Limitations and conclusions

Limitations. Three limitations qualify the map. First, the gain baseline is a documented partial reconstruction: it uses the magnetized Harris susceptibilities and warm-fluid eigenmode projection with the standard CBET gain coefficient, but omits the magnetization correction to the ponderomotive drive (assumption (i)), so the high-field boundary is partly a statement about that omission. Second, the hydrogen cells carry no documented uncertainty band and are reported as illustrative only. Third, the new intensity-sweep runs use a reduced spatial scale and resolution; the reduction fails the pre-registered absolute-gain tolerance against the published reference point (the gain is transient and length-limited, so the shorter domain under-measures it), so those runs are used only for the intensity trend and the field ordering, and the intensity boundary is correspondingly a region rather than a sharp line.

Conclusions. We have mapped where the linear kinetic theory of magnetized CBET remains predictive against the published PIC data and new reduced-scale simulations, using a pre-registered agreement statistic and boundary rule. The predictive region is compact and physically organized: the field strength and the field azimuth carry the agreement signal, the theory holds at near-parallel field below ~ 10 MG, and it breaks through two distinct mechanisms — laser-drive/eigenmode magnetization at high field, and magnetized-IAW (finite-Larmor-radius) physics at near-perpendicular field — with ion nonlinearity (trapping) setting the intensity boundary near the upper swept range. The map gives ICF practitioners a concrete rule for when the fast Vlasov surrogate can replace a kinetic simulation, and identifies the specific theory approximations whose extension would most expand its reliable domain.

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